

Prediction of disulfide-bonded cysteines in proteomes with a hidden neural network

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A hidden neural network-based method is used to predict the bonding state of cysteines starting from the residue sequence of the protein chain. The method scores as high as 89% and 86% per cysteine residue and per protein, respectively, and in this overcomes other predictors of the same category. We then explore the efficacy of our predictor in computing the disulfide content of the whole proteome of *Escherichia coli* (K12 and O157), *Aeropyrum pernix*, *Thermotoga maritima*, and *Homo sapiens*. We find that the percentage of extracellular disulfide containing proteins is higher than that of intracellular one, and that the human proteome is by far the one with the highest content of sulfur-sulfur linkages in proteins.

Keywords: Disulfide bond / Disulfide proteome / Hidden Markov models / Neural networks / Sequence analysis

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1 Introduction

The formation of disulfide bonds between paired cysteine residues is a key step in the folding process of many proteins [1]. The post-translational modification promoting the sulfur-sulfur chemical linkage of nonadjacent (in most cases) cysteines in the protein chains is performed both in yeasts and bacteria by enzymatic systems in extracytoplasmic compartments [2, 3]. Therefore, proteins containing stable disulfide bonds are rarely found in the cytoplasmic compartments of any organisms [2, 3]. In bacteria they are located in extracytoplasmic compartments or secreted into the external medium; in eukaryotic cells they are to be found in compartments such as the plasma membrane, the endoplasmic reticulum or secreted into the external medium [4]. Certain archaea, however, seem to be an exception to this rule, since it has been reported that they are endowed with a large number of disulfide-bonded proteins in the cytoplasm [5].

Two classes of disulfide-bonded proteins are known: the first in which the inter-cysteine linkage is a stable part of the final folded structure and the second in which cysteine pairs alternate between the reduced and the oxidized

states. In several organisms, ranging from *Escherichia coli* to humans, disulfide bond formation is important for the proper folding of secreted proteins, including bacterial virulence factors and mammalian secreted glycoproteins [2, 3]. Also the redox cycling of the disulfide bond may be central to an enzyme activity or to protein activation [4]. In the genomic era, when the proteomes of hundreds of different organisms from bacteria to humans are known, it is particularly interesting to be able to highlight the “disulfide proteome” for the comprehensive analysis of the cell function and regulation [6].

In this paper, we modify and refine a hidden neural network-based predictor for computing the probability of a cysteine residue to be involved in a disulfide bond starting from the protein sequence. The new predictor scores as high as 86% in correctly predicting the bonding state of cysteines per protein of a training set containing 1631 chains from different sources. With this tool we then highlight the disulfide proteome content of different organisms from bacteria to man and find that the predicted human disulfide proteome is larger than that of bacteria and archaea.

2 Materials and methods

2.1 Databases

We start by deriving a selected set of nonredundant sequences (sequence identity < 25%) from July 2003 release of the PDB database of known protein structures

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Abbreviation: HMM, hidden Markov model

(available at www.biocomp.unibo.it/CysDatabase.txt). Our sets comprise 1631 chains of known structure for a total sum of 412049 residues. 436 proteins contain disulfide bonds, and 280 chains are endowed with more than one disulfide bond per chain. 6961 cysteine residues are present in the set. The total number of disulfide-bonded cysteines is 2308 (33%), and the number of free cysteines is 4653 (67%). To train the neural network, we use a 27 residue wide window centered on each cysteine residue, as extracted from the protein set. In total 187 947 residues are presented to the network. Our basic requirement for training and testing the predictors is to have a nonredundant set of structures. With this constraint, the set includes 1631 proteins with at least one cysteine and known with atomic resolution. If we consider the corresponding SWISS-PROT annotated files, we find that out of the set, 320 proteins are labeled with the key oxidoreductase (20%) and among these only 16 chains contain cysteines that are annotated as redox active (about 1%); also 88 proteins have cysteines coordinating metals (5%). Due to the paucity of examples at hand for the different cysteine functional types and considering the vast majority of cases, our choice is to classify all the cysteines in the set as disulfide-bonded and free, without discriminating within the different functional types. Cysteines coordinating metals are classified free. Also, within the disulfide-bonded cysteines, only 6% are derived from oxido reductase proteins. Proteomes were taken from the publicly available database at NCBI (www.ncbi.nlm.nih.gov/Genomes) in August 2003. The human proteome was downloaded from Ensembl (release 15.33.1, www.ensembl.org).

2.2 Hidden neural network

This type of predictor has been described before [7] and it overperforms other predictors developed for the same purpose [8–13]. Briefly, the hidden neural network is a combined system of neural networks and hidden Markov models (HMMs), where the network output is the state emission probability of the probabilistic model (HMM). It has been proved and discussed that by this both local and global characteristics conducive to the disulfide bridge formation in a protein chain are better captured by the predictive system and that the performance greatly improves over simple neural networks system or other statistical methods [7]. In the present paper, we implement a new version of the hidden neural network system (CYSPRED2), where a classical feed-forward neural network is combined with a seven state probabilistic model, discriminating free cysteines from disulfide-bonded ones (Fig. 1). The seven states of the HMM explicitly model the three possible classes that are taken into account

for each protein: without disulfide bonds, with all the cysteines involved in disulfide bonds and with disulfide-bonded and free cysteines in the same chain (see Fig. 1). The formalism describing our procedure is detailed elsewhere [7]. The neural network is a classical feed-forward neural network trained with the back-propagation algorithm, using sequence profile as input. This is derived from multiple sequence alignment [7], using as input a sequence profile, generated with PSIBLAST [14] on the nonredundant dataset of sequences (the profile is computed with the blastpgp program endowed with the -Q option set to collect the profiles, and three rounds of PSIBLAST with an e-value of 0.001). A 20-fold cross validation procedure is adopted for testing the system and the scoring indexes are average values. CYSPRED2 will be available on the web at www.biocomp.unibo.it.

2.3 Other predictors

In this paper, we use two other predictors that we previously implemented. The topology of the disulfide bridges is predicted with DCON [15, 16]. This predictor computes the connectivity of cysteine residues by equating this problem to that of finding the graph matching with the maximum weight. DCON takes advantage of a neural network to predict the cysteine pairwise interaction and scores as high as 66% when two bridges (four paired cysteines) are present in the chain. Unfortunately the performance of the predictor is decreasing at increasing number of disulfide bridges in the protein. HUNTER [17] is a suite of programs that combines several predictors, including a signal peptide predictor [18], a HMM-based predictor specific for the all-beta outer membrane proteins [19], and a predictor specific for all-alpha inner membrane proteins [20]. HUNTER, on the basis of structure prediction, discriminates globular from membrane proteins, and among these, outer (all-beta) from inner (all-alpha) membrane proteins in bacteria. A modified version of Hunter (ManHUNTER), among membrane proteins, predicts beta-barrel membrane proteins (mainly voltage-dependent anion channels, referred to as human porins) present in the outer mitochondrial membrane [21].

3 Results and discussion

3.1 From sequence to disulfide bond topology: the state of the art

With a sequence at hand, what can we predict in terms of the structural properties of cysteine residues? This question is relevant to the protein folding pathway, since knowing whether (and which) cysteine residues form disulfide

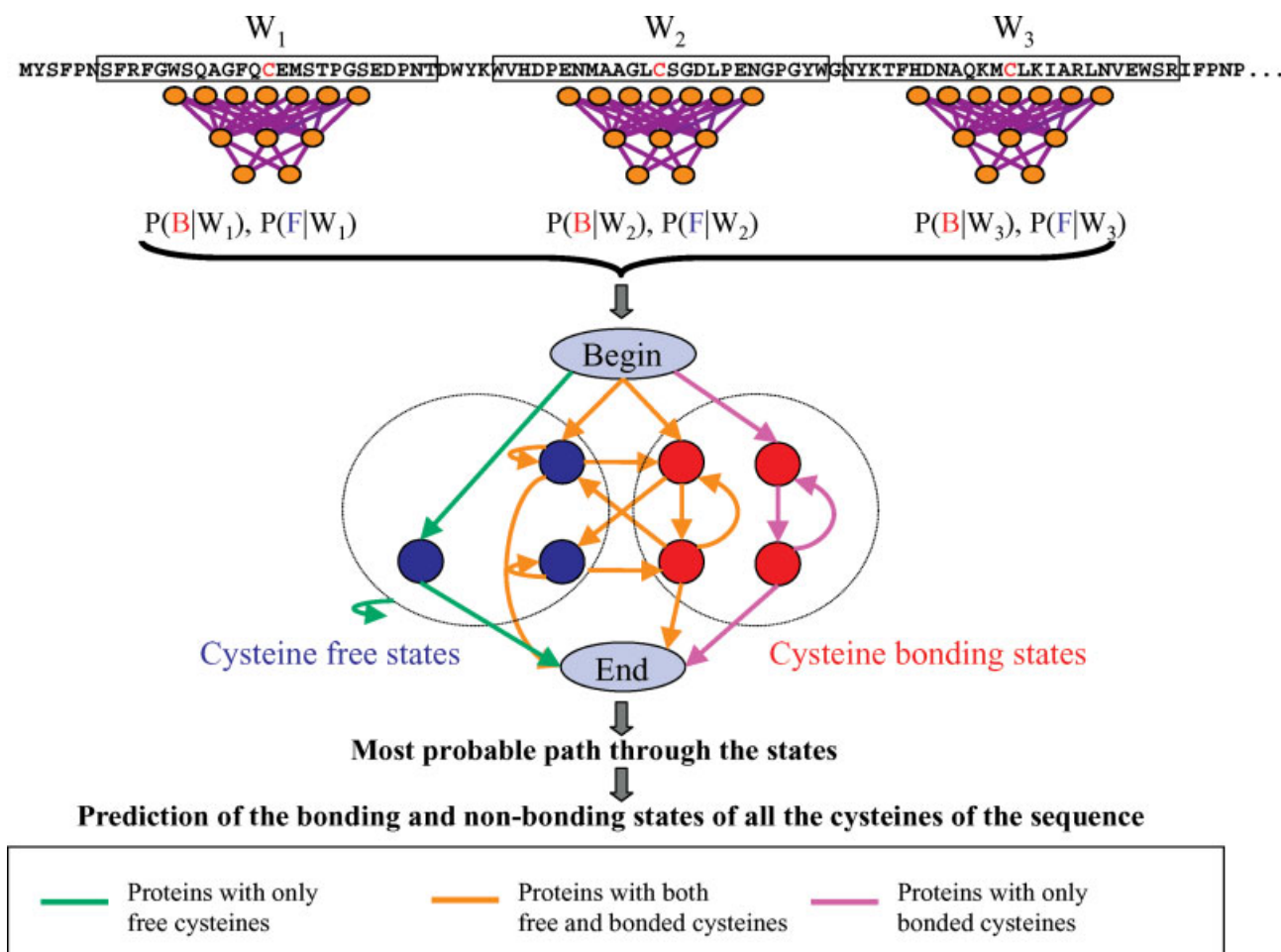


Figure 1. Scheme of the hybrid system comprising a feed forward neural network and a HMM (hidden neural network). The first predictor consists of a neural network (NN) that uses evolutionary information as input and is trained on a 27 long window centered on the cysteine residues of the protein chain. The local context information is conducive to the bonding or nonbonding state of the cysteine residue [7]. Then the NN outputs are used as emission probabilities of a seven-state HMM that constrains the number of cysteines predicted in the disulfide bonding state per chain to be even. The arrows represent the allowed transitions. The red and blue balls represent the bonding and nonbonding cysteine states, respectively. The path can end only from an even state. This guarantees that only correct even predictions are assigned when considering intra-chain disulfide bonds. Inter-chain disulfide-bridged cysteines are considered as free cysteines, since they represent only 0.8% of the training set of the cysteine containing segments. $P(B|W_i)$ and $P(F|W_i)$ are the probabilities for the Bonding (B) and Free (nonbonding, F) state, respectively, at a given local context (W).

bonds may help in unraveling which steps will occur during the folding process and pose constraints to the three-dimensional architecture. In spite of its relevance, the prediction of the cysteine bonding states has been rarely addressed [7–12] and only very few predictors are available [13].

We recently introduced a combined system of neural networks and HMMs [7], capable of predicting the bonding state of cysteine in proteins, starting from the residue sequence with accuracy per protein of 84%, largely supe-

rior to that previously achieved. With this approach both local and global information are considered in order to predict the bonding state of a cysteine residue in a given sequence environment. In Fig. 1, a recently modified version of this predictor (CYS-PRED2) is shown. Here, we perform global probabilistic modeling of the protein sequence with an HMM that includes seven states, to be compared with the four ones we introduced before [7]. In this case, the discrimination between bonded and unbonded (free) cysteine is computed by considering three different protein classes that the model explicitly

takes into consideration: each chain may contain only free, disulfide-bonded, or both disulfide-bonded and free cysteines. The network output is, as before [7], used as the state emission probability value of the HMM. The results (Table 1) are obtained by adopting a 20-fold cross validation procedure. The statistical indexes adopted to score the method are explained in the footnote of Table 1. Previously [7], with a four-state HMM model to discriminate the same three protein classes, we obtained a lower score (accuracy per protein is now 86 vs. the previous 84% [7]). Considering also that the other statistical indexes shown in Table 1 are now slightly higher than those previously published, we conclude that the hidden neural networks system endowed with the seven-state HMM is somewhat better in performing the task at hand.

Table 1. Performance of the hidden neural network-based predictor (20-fold cross validation)

Q2	C	Q(B)	Q(F)	P(B)	P(F)	Q2 _{prot}
89%	0.76	75%	95%	92%	89%	86%

B, cysteine bonding state; F, cysteine free state; Q2, number of correctly predicted cysteines/total number of cysteines; C, correlation index; Q(class), number of cysteines correctly predicted in *class*/number of cysteines in *class*; P(class), number of cysteines correctly predicted in *class*/number of cysteines predicted in *class*; Q2_{prot}, number of proteins correctly predicted/ total number of proteins.

An example of the predictor capability is shown in Fig. 2. The target sequence is the gamma-chymotrypsin chain from *Bos taurus*. With CYPRED2, each cysteine in the chain is predicted to be in a bonding or free state. The chain contains five cysteines that are correctly predicted to be either in the bonding or nonbonding state.

A more difficult task is the prediction of the topology of the disulfide bridges in a protein. While disulfide connectivity in proteins can be classified [22], the prediction of the connectivity of disulfide bridges (which cysteine is bonded to which) in the protein chain is extremely difficult, due to the paucity of examples known with atomic resolution [15]. For proteins with two bridges in the chain a neural network-including predictor can score as high as 66% [16]. However, this performance decreases at increasing numbers of bridges per protein. If we consider the previous example, after predicting the bonding state of cysteines, a second predictor (DCON, available at www.biocomp.unibo.it) can compute the connectivity of the cysteines involved into a disulfide bridge (which cysteine is bound to which in the correct protein folding). In this particular case, none of the predictors include the

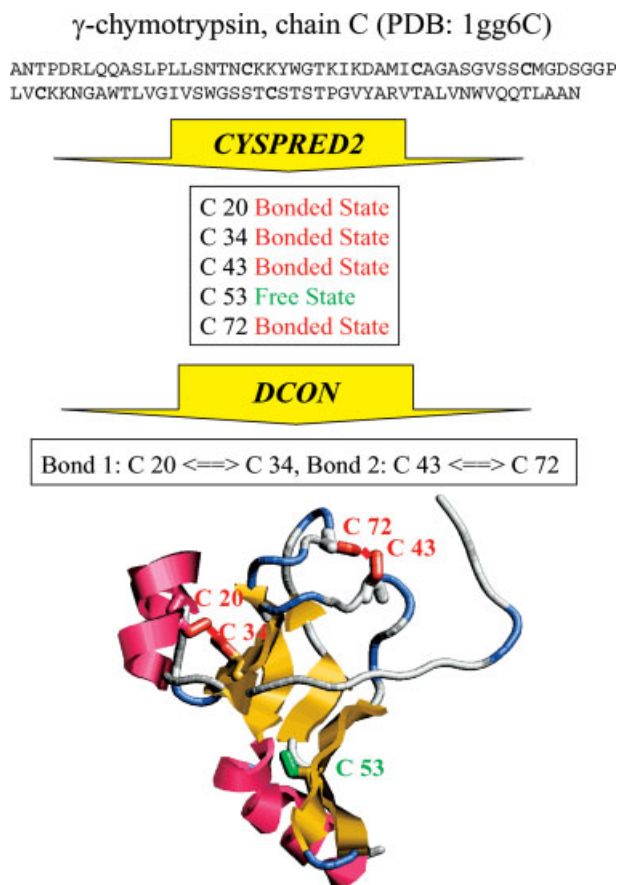


Figure 2. Prediction of bonding state and connectivity pattern of cysteine residues in gamma-chymotrypsin, chain C (PDB: 1gg6C). CYPRED2 predicts the bonding state and DCON predicts the connectivity pattern of the four cysteines in the bonding state.

protein chain in the training/testing set and predictions match exactly expectations, as shown. However, while CYPRED2 has a high accuracy in predicting the disulfide-bonded state, DCON in its present version is moderately satisfactory only for proteins containing two disulfide bridges, and therefore is not suited to address proteome-wide analysis, searching for connectivity patterns. More examples are necessary in order to implement a predictor with higher efficiency.

3.2 Disulfide proteome of different organisms

It is particularly interesting to be able to predict the entire content of disulfide rich proteins in a proteome (the disulfide proteome). This may help in different considerations and/or planning of proteomic experiments and eventually can be related to experimental determinations. Our goal is therefore to generate a list of putative proteins in different proteomes that according to our analysis are character-

Table 2. Predicting the cysteine bonding state in proteomes

Organism	No. of proteins	No. of proteins with bonded cysteines	No. of cysteines	No. of bonded cysteines	No. of free cysteines	No. of beta-barrel membrane proteins with bonded cysteines ^{a)}	No. of all-alpha membrane proteins with bonded cysteines ^{a)}	No. of globular proteins with bonded cysteines ^{a)}
<i>E.coli</i> K12	4,173	247 (6%)	15 380	976 (6%)	14 404 (94%)	17 [65]	26 [907]	204 [3201]
<i>E.coli</i> O157	5,361	369 (7%)	18 934	1 438 (8%)	17 496 (92%)	8 [78]	42 [1034]	319 [4249]
<i>Thermotoga maritima</i>	1,846	34 (2%)	4 124	118 (3%)	4 006 (97%)	1 [18]	5 [370]	28 [1458]
<i>Aeropyrum pernix</i>	1,841	134 (7%)	3 995	460 (12%)	3 535 (88%)	–	27 [315]	107 [1526]
<i>Homo sapiens</i>	32,035	6780 (21%)	345 081	83 922 (24%)	261 159 (76%)	346 [2470]	1 190 [6031]	5 244 [23534]

a) Figures within square brackets are the total number of proteins with free and bound cysteines in the structural class.

ized by the presence of disulfide bonds. We adopt a two-step procedure that takes advantage also of another predictor (HUNTER, [17]). First each of the proteomes is filtered with a procedure implemented in-house, according to which a suite of programs can discriminate globular from membrane proteins, and among these, outer (all-beta) from inner (all-alpha) membrane proteins in microbes on the basis of structure prediction. A modified version of HUNTER (ManHUNTER) predicts also beta-barrel membrane proteins of eukaryotes. After that, each protein structural class is then predicted with CYPRED2 that discriminates disulfide containing from disulfide noncontaining proteins in the different organisms. For sake of generalization here we consider two bacteria, *Escherichia coli* wild type (K12) and its pathogenic strain (O157), the hyperthermophile bacterium *Thermotoga maritima*, the hyperthermophile archaeon *Aeropyrum pernix*, and the human proteome.

The results are shown in Table 2. From our prediction it appears that among the different species, *Homo sapiens* is the one characterized by the highest percentage of proteins with disulfide bonds (21%). Also, the number of proteins with disulfide bonds increases in the pathogenic form of *E. coli* as compared with the wild type. As to hyperthermophiles, the bacterium is endowed with less disulfide containing proteins than the archaeon, whose content in turn is similar to that of the pathogenic form of *E. coli*.

3.3 How reliable are the proteome-wide predictions?

An important question is to which extent are the results shown in Table 2 reliable. In order to answer an estimate of the rate of false positives and negatives is at order.

One should first of all take into consideration that our method relies on the application of two different predictors: HUNTER and CYPRED2. HUNTER discriminates different protein classes based on the structural prediction. In order to assess whether HUNTER has a good discriminative capability we compared the predictions for each sequence in the proteome with the SWISS-PROT annotation when available. This, on a large scale, was possible only in the case of *E. coli* K12 proteome, with more than 70% of the chains well or partially well annotated in the SWISS-PROT database [17]. We computed the accuracy of the prediction for all-beta membrane proteins: the rate of false negatives (proteins that are missed in the prediction) and of false positives (proteins that are falsely predicted in the category) is 20 and 10%, respectively [17]. In the same bacterium, rates of false negatives and positives are equal for all-alpha membrane proteins (9%); also, we found equal values for globular proteins (5%). For other proteomes, including the ones here analyzed, the percentage of proteins well annotated is much less and this hampers the application of the same procedure. However, HUNTER was trained/tested using only a nonredundant set of PDB structures, from different sources, and its performance can therefore be assumed to be reasonably independent of the particular proteome analyzed. With this in mind, we should consider that our predictor tends to underestimate more than overestimate the number of all-beta membrane proteins; for the remaining two other structural classes discriminated, underestimation and overestimation have reasonably low equal values.

The rate of false positive and false negative classification of CYPRED2 is 8% and 25%, as evaluated by considering (1-Q(B)) and (1-P(B)) in Table 1. This suggests that the predictor may act more as an under-predictor (some true bonded cysteines may be missed) than as an

over-predictor. With this scenario, one should consider that numbers in Table 2 may be an estimate of the actual numbers. For example, when considering the prediction of disulfide bonds in the outer membrane protein class (Table 2), the figure is likely to be an underestimate of an already underestimated set of chains. However, the rate of false positives and negatives that we computed is the same independently of the organisms (as discussed above) and therefore the relative abundance of cysteine containing proteins in the different organisms should be considered significant.

3.4 More disulfide bonds in the cytoplasmic or extracytoplasmic proteins?

We can assume that a protein is secreted in the extracytoplasmic region when its sequence is endowed with a signal peptide. In Table 3 the disulfide containing proteins of the globular type are listed as a function of their location, as labeled by the presence or not of a predicted signal peptide. For this, we made use of a signal peptide predictor also implemented in-house [18]. If we consider our definition of cytoplasmic *versus* extracytoplasmic protein (depending on the absence or presence of a signal peptide), we can conclude that the number of secreted proteins is a small fraction of the proteome. The absolute number of proteins predicted with disulfide bonds is higher in the cytoplasmic region as compared with the extracytoplasmic one. This is apparently in contrast with the common belief that intracellular disulfide bonds are rare due to the reducing environment, or at least that many proteins catalyzing redox processes during catalytic cycles form transient and unstable disulfide bridges [2, 3]. However, it should be also considered that the num-

ber of secreted proteins is much smaller than the number of cytoplasmic proteins and then that the common belief holds if we consider the percent values in Table 3.

Our predictor has been trained/tested on a dataset comprising cysteines in a disulfide-bonded state out of about 80% of proteins with stable disulfide bonds and about 20% of redox-active proteins. The division that we made took only the property of being disulfide bonded or free into consideration, irrespectively of the functional role of the cysteine residue (see also Section 2). For this reason, the predictor generalizes over local patterns conducive to bond formation, rather irrespectively of the protein function. Indeed among globular cytoplasmic proteins predicted with disulfide bonds and also annotated in the SWISS-PROT database, some 10% in the different organisms belong to the redox-active proteins. A large fraction of proteins (Table 3) is however not annotated (this fraction is higher for the cytoplasmic proteins than for the extracytoplasmic ones) and for this reason it is presently difficult to postulate a role for the disulfide bond, if present.

Our data indicate that 5% of the cytoplasmic globular proteins of *E. coli* have disulfide bonds. Albeit with a completely different computational procedure based on building by homology, the same amount was estimated before [5]. Similarly to what previously reported [5], we also find that this figure increases in *Aeropyrum pernix*, a hyperthermophilic archaeon, although to a much less extent (10% vs. 40% [5]). From the data shown in Table 3, we can however conclude that the percentage of proteins with disulfide bonds is higher in the extracytoplasmic compartment, confirming that in this region proteins are more likely to form disulfide bridges due to the oxidizing and sometimes disruptive environment [23].

Table 3. Prediction of bonded cysteines in cytoplasmic and extracytoplasmic globular proteins

Organism	Cytoplasmic Globular ^{a)}				Extracytoplasmic Globular ^{a)}			
	No. of proteins	No. of cysteines	No. of proteins with bonded cysteines	No. of bonded cysteines	No. of proteins	No. of cysteines	No. of proteins with bonded cysteines	No. of bonded cysteines
<i>E. coli</i> K12	2 796	11 125	131 (5%) [82]	534 (5%)	405	993	73 (18%) [24]	272 (27%)
<i>E. coli</i> O157	3 715	13 891	241 (6%) [188]	972 (7%)	534	1 389	78 (15%) [46]	260 (19%)
<i>Thermotoga maritima</i>	1 346	3 546	21 (2%) [15]	84 (4%)	112	164	7 (6%) [7]	16 (10%)
<i>Aeropyrum pernix</i>	1 254	2 784	80 (6%) [78]	268 (10%)	62	129	13 (21%) [12]	34 (26%)
<i>Homo sapiens</i>	20 422	204 204	4 086 (20%) [2113]	42 322 (21%)	3112	41 548	1158 (37%) [159]	20 000 (48%)

The protein in the set is predicted with (a) or without (b) signal peptide.

Within square brackets is the number of proteins without annotation in the SWISS-PROT database.

4 Concluding remarks

In this paper, we address the issue of predicting the disulfide proteome of different species using a system specifically implemented for computing the probability that a cysteine residue in a protein chain may or may not be involved in a sulfur-sulfur linkage. *Homo sapiens* is by far the organism with the highest percentage of disulfide containing proteins when compared to archaea and bacteria and with the highest percentage of extracellular proteins with disulfide bonds. We propose that our computational procedure can be used to unravel relevant features of protein function and structure.

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5 References

- [1] Wedemeyer, W. J., Welker, E., Narayan, M., Scheraga, H. A., *Biochemistry* 2000, 39, 4208–4216.
- [2] Hogg, P. J., *Trends Biochem. Sci.* 2003, 28, 210–214.
- [3] Kadokura, H., Katzen, F., Beckwith, J., *Annu. Rev. Biochem.* 2003, 72, 111–135.
- [4] Hiniker, A., Bardwell, J. C., *Biochemistry* 2003, 42, 1179–1185.
- [5] Mallick, P., Boutz, D. R., Eisenberg, D., Yeates, T. O., *Proc. Natl. Acad. Sci. USA* 2002, 99, 9679–9684.
- [6] Yano, H., Kuroda, S., Buchanan, B. B., *Proteomics* 2002, 2, 1090–1096.
- [7] Martelli, P. L., Fariselli, P., Malaguti, L., Casadio, R., *Prot. Eng.* 2002, 15, 951–953.
- [8] Muskal, S. M., Holbrook, R. S., Kim, S. H., *Prot. Eng.* 1990, 3, 667–672.
- [9] Fiser, A., Cserzo, M., Tudos, E., Simon, I., *FEBS Lett.* 1992, 302, 117–120.
- [10] Fariselli, P., Riccobelli, P., Casadio, R., *Proteins* 1999, 36, 340–346.
- [11] Fiser, A., Simon, I., *Bioinformatics* 200, 6, 251–256.
- [12] Mucchielli-Giorgi, M. H., Hazout, S., Tuffery, P., *Proteins* 2002, 46, 243–249.
- [13] Dosztanyi, Z., Magyar, C., Tusnady, G. E., Cserzo, M. *et al.*, *Nucleic Acids Res.* 2003, 31, 3359–3363.
- [14] Altschul, S. F., Madden, T. L., Schaffer, A. A., Zhang, J. *et al.*, *Nucleic Acids Res.* 1997, 25, 3389–3402.
- [15] Fariselli, P., Casadio, R., *Bioinformatics* 2001, 17, 957–964.
- [16] Fariselli, P., Martelli, P. L., Casadio, R., in: Daminai, E., Howlett, R. J., Jain, L. C. (Eds.), *Knowledge-based Information Engineering Systems and Allied Technologies*, IOS Press, Amsterdam 2002, pp. 464–476.
- [17] Casadio, R., Fariselli, P., Finocchiaro, G., Martelli, P. L., *Protein Sci.* 2003, 12, 1158–1168.
- [18] Finocchiaro, G., Fariselli, P., Casadio, R., *Bioinformatics* 2003, 19, 2498–2499.
- [19] Martelli, P. L., Fariselli, P., Krogh, A., Casadio, R., *Bioinformatics* 2002, 18, Suppl. 1, S46–53.
- [20] Martelli, P. L., Fariselli, P., Casadio, R., *Bioinformatics* 2003, 19, Suppl. 1, I205–I211.
- [21] Casadio, R., Jacoboni, I., Messina, A., De Pinto, V., *FEBS Lett.* 2002, 520, 1–527.
- [22] Harrison, P. M., Sternberg, M. J. E., *J. Mol. Biol.* 1994, 244, 448–463.
- [23] Thornton, J. M., *J. Mol. Biol.* 1981, 151, 261–287.